

Chapter 3: Functional group protection

General considerations

- Main features of a protecting group

- Compatibility and orthogonality between protecting groups

Classification of protecting groups according to the cleavage conditions

Functional group protection

- Protection of hydroxyl, carbonyl, carboxyl and amino groups

References:

- ✓ F.A. Carey, R.J. Sundberg, *Advanced Organic Chemistry. Part B: Reactions and Synthesis*, 4^a ed., Kluwer, 2001. Capítol 13.1
- ✓ T.W. Greene, P.G.M. Wuts, *Protective Groups in Organic Synthesis*, 3^a ed., Wiley, 1999
- ✓ P.J. Kociński, *Protecting Groups*, Thieme, 1994.
- ✓ J. Robertson, *Protecting Group Chemistry*, Oxford Chemistry Primers, 2000

Chapter 3. Functional group protection

3.1. General considerations

- Issues that define the suitability of a protecting group:
 - a) It should be cheap or readily available
 - b) It should be easily and efficiently introduced
 - c) It should be easy to characterize and avoid such complications as the creation of new stereogenic centers
 - d) It should be stable to the widest possible range of reaction conditions and to the work-up steps
 - e) It should be stable to the different separation and purification techniques, such as chromatography
 - f) It should be removed selectively and efficiently under highly specific conditions
 - g) The by-products of the deprotection should be easily separated from the substrate

- Two protecting groups are compatible if they show a gradual lability towards a given reagent.

- Two protecting groups are orthogonal if one can be removed in the presence of the other and in any order. Their removal mechanisms are different.

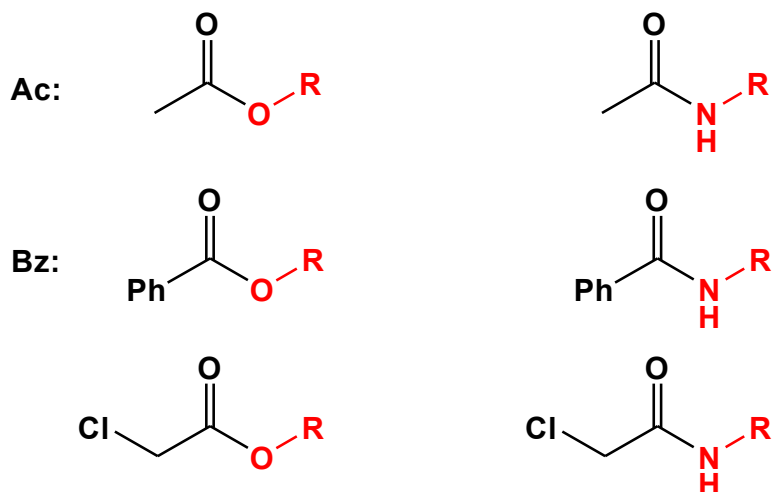
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3.2. Classification of protecting groups according to the cleavage conditions

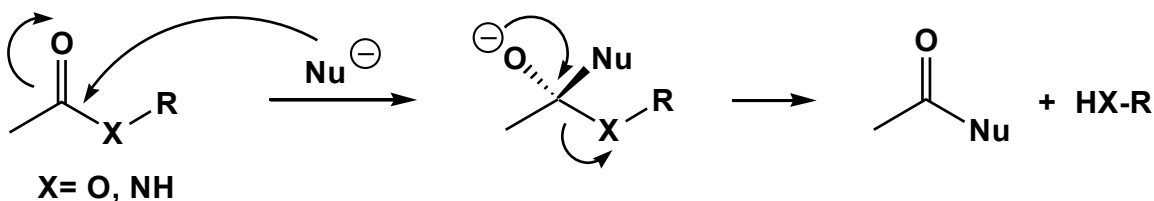
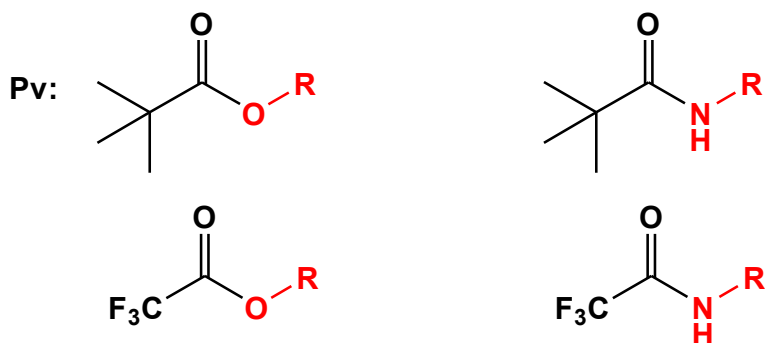
- 3.2.1. Protecting groups cleaved by basic solvolysis
- 3.2.2. Protecting groups cleaved by acid
- 3.2.3. Protecting groups cleaved by a heavy metal
- 3.2.4. Protecting groups cleaved by fluoride ions
- 3.2.5. Protecting groups cleaved by reductive elimination
- 3.2.6. Protecting groups cleaved by β -elimination reactions
- 3.2.7. Protecting groups cleaved by hydrogenolysis
- 3.2.8. Protecting groups cleaved by oxidation
- 3.2.9. Protecting groups cleaved by dissolving metal reduction
- 3.2.10. Protecting groups cleaved by transition metal catalysis
- 3.2.11. Protecting groups cleaved by light

3.2.1. Protecting groups cleaved by basic solvolysis

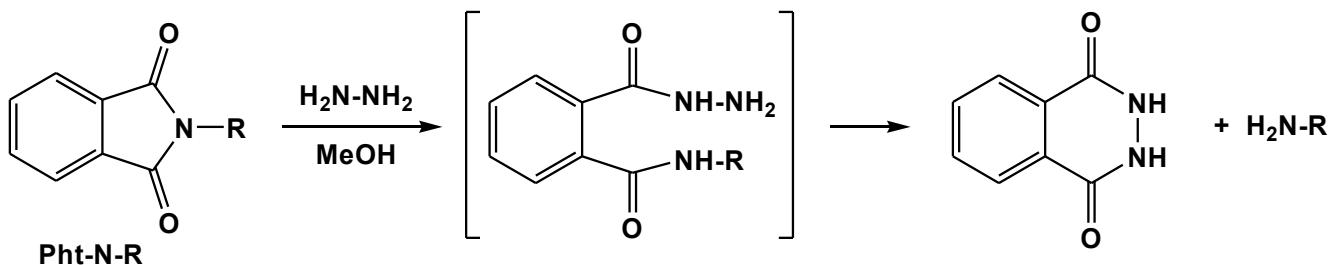
Acyl derivatives of hydroxyl and amino groups:



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Phthaloyl group, a special case:



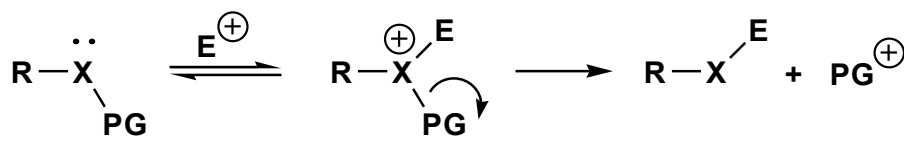
3.2.2. Protecting groups cleaved by acid

They can be divided into two subsets:

- Tertiary alkyl or benzylic amines, ethers, esters, carbonates and carbamates
- O,O-acetals

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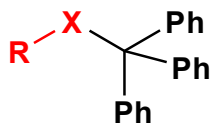
- Tertiary alkyl or benzylic amines, ethers, esters, carbonates and carbamates



X = O, NH, S

E⁺ = H⁺ or Lewis acid

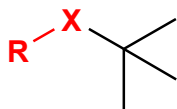
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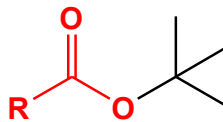
X = O, NH, S

Labile to AcOH

tBu:

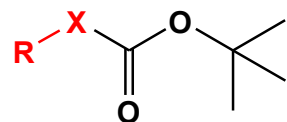


X = O, NH



Labile to TFA

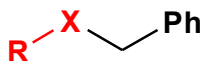
Boc:



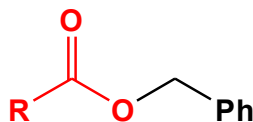
X = O, NH

Labile to TFA

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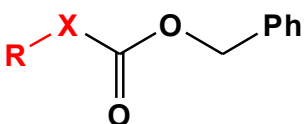


X = O, NH, S



Labile to HBr or HF

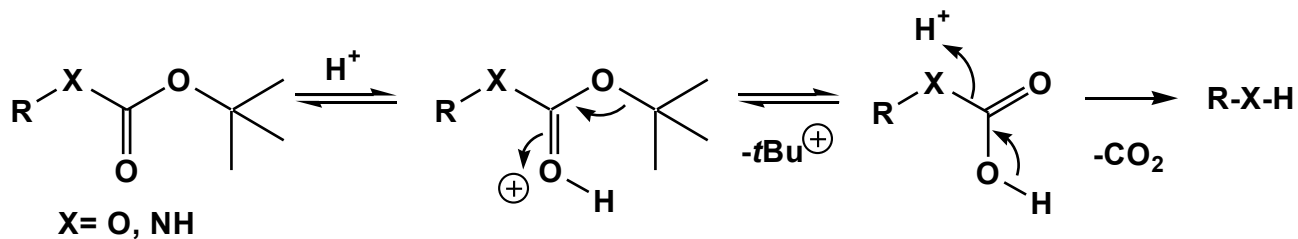
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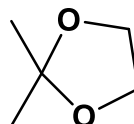
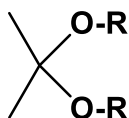
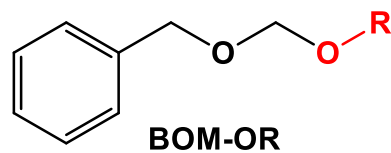
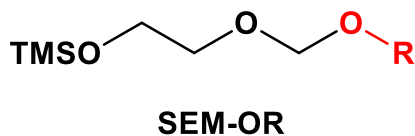
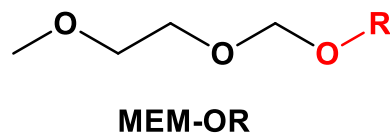
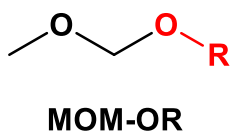
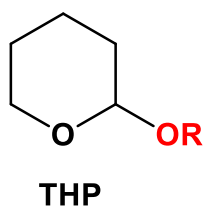
X = O, NH

Labile to HBr or HF

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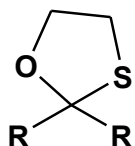
➤ O,O-acetals



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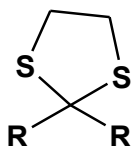
3.2.3. Protecting groups cleaved by a heavy metal

O,S-acetals

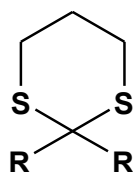


oxathiolane

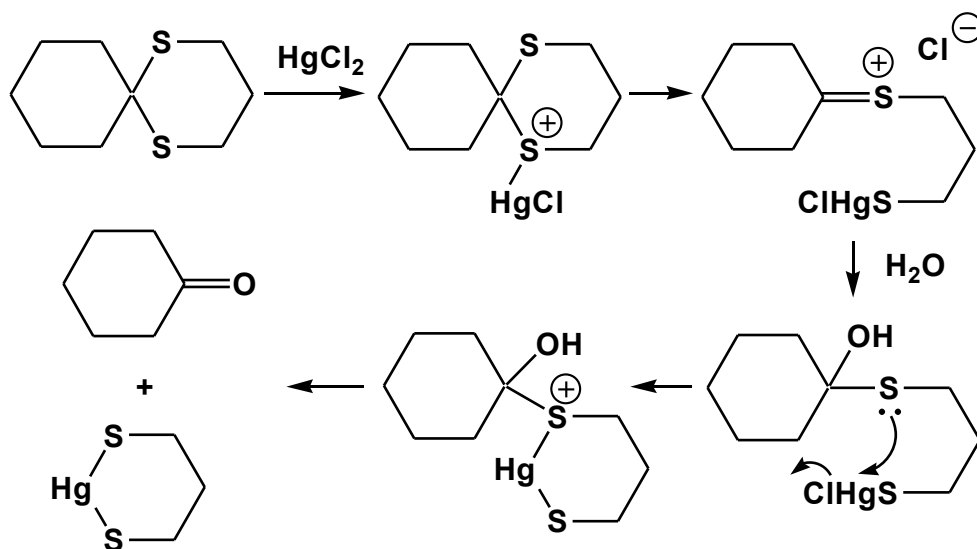
S,S-acetals



dithiolane



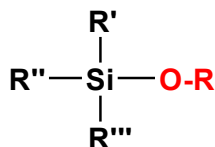
dithiane



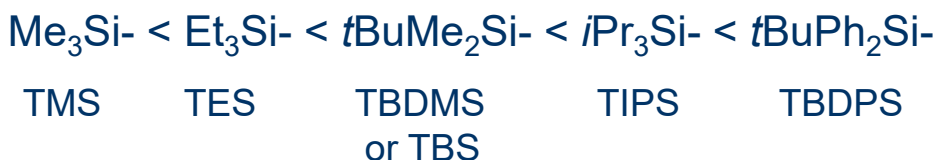
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3.2.4. Protecting groups cleaved by fluoride ions

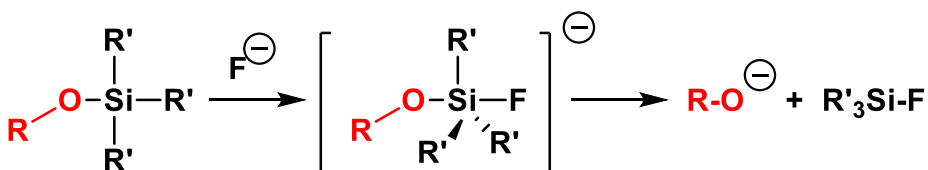
➤ Cleavage of Si-O bonds: silyl ethers



Acid catalyzed
solvolysis:

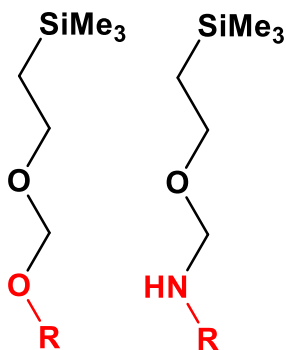
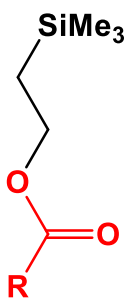


Base catalyzed
solvolysis:



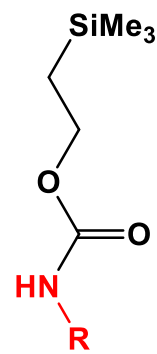
Reagents: $n\text{Bu}_4\text{N}^+\text{F}^-$ (TBAF) in THF or HF in MeCN

➤ Cleavage of Si-C bonds



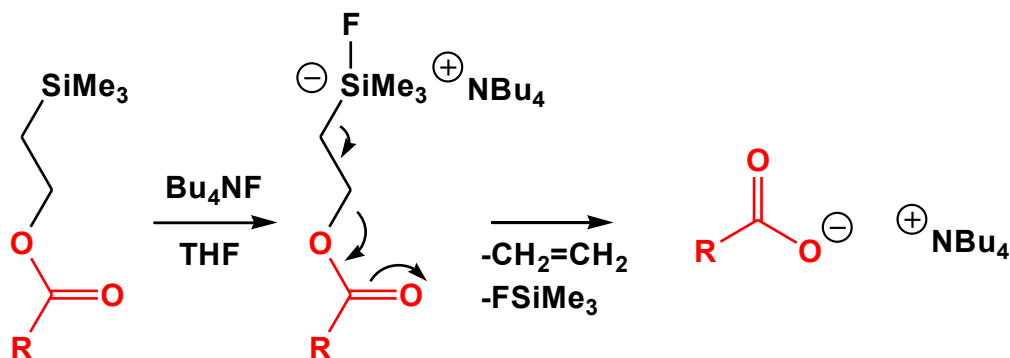
SEM

TMSEC



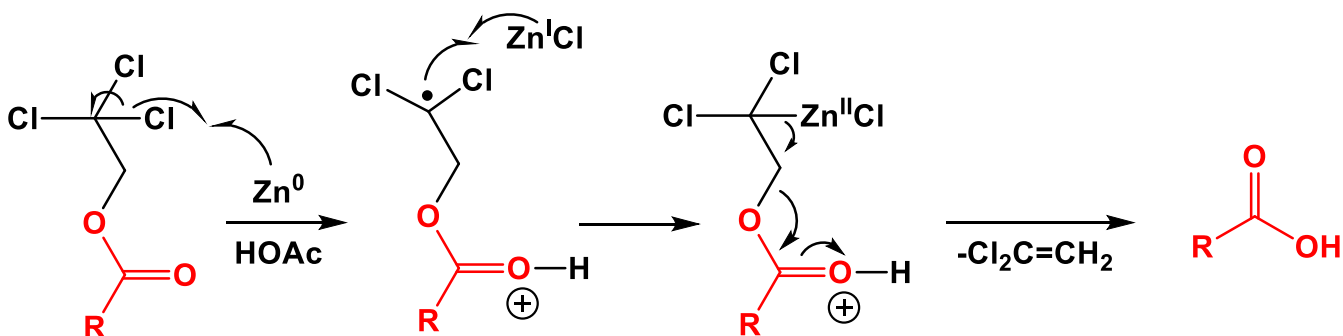
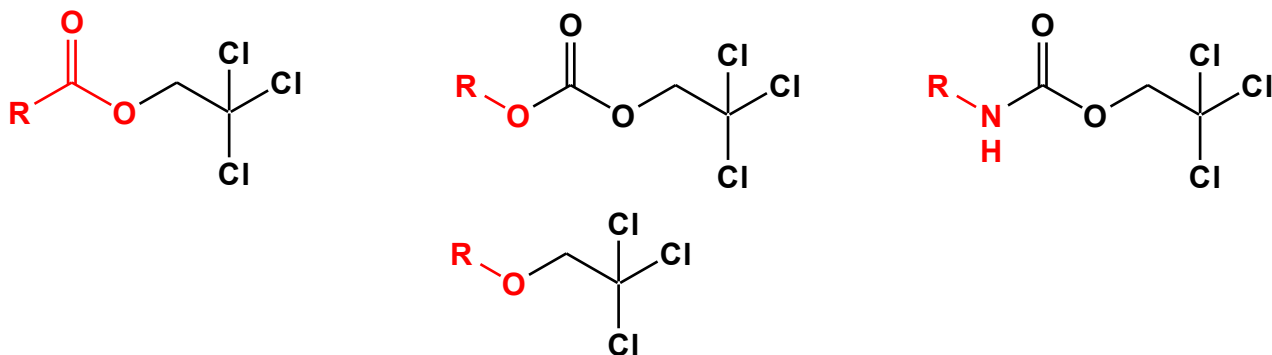
Teoc

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3.2.5. Protecting groups cleaved by reductive elimination

2,2,2-Trichloroethyl derivatives:

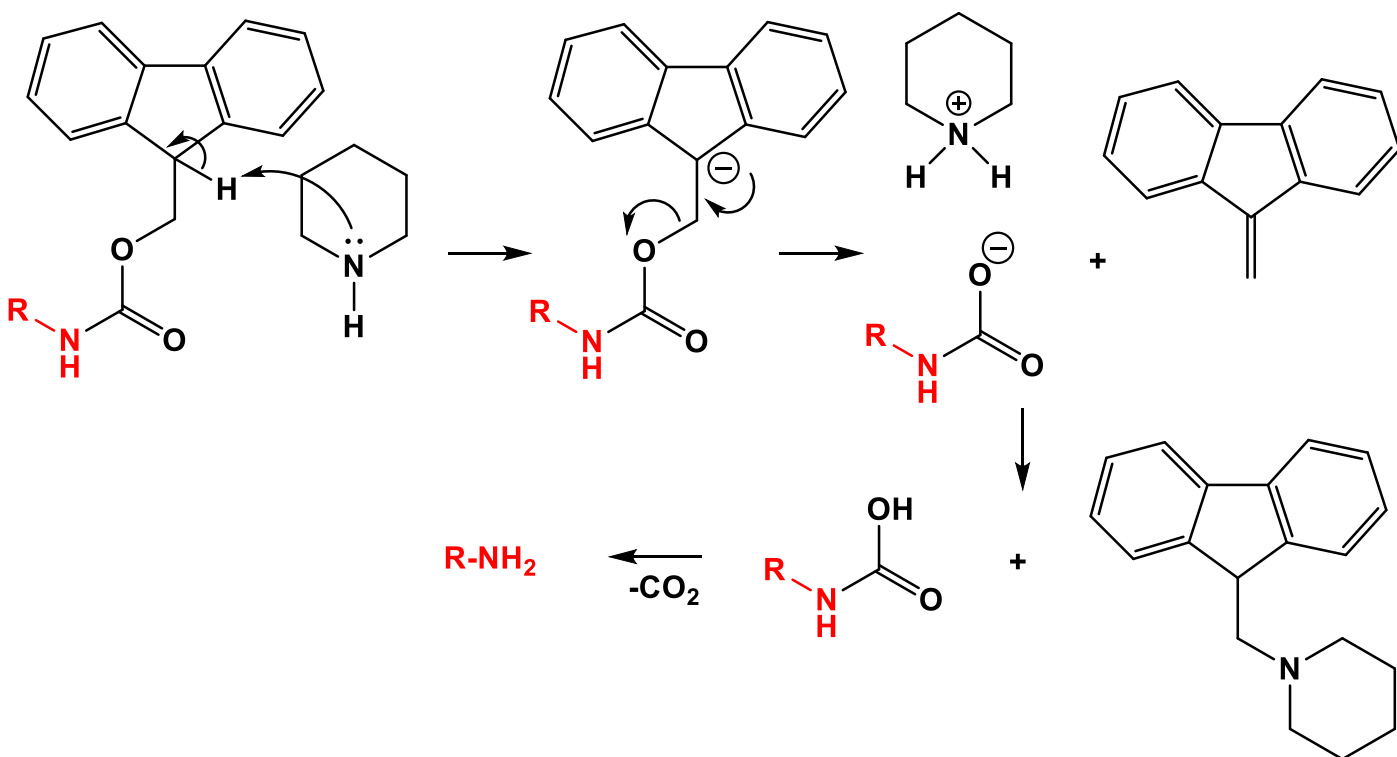


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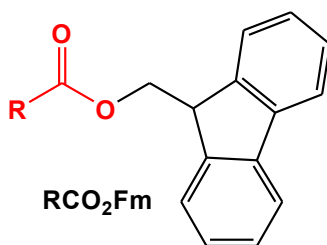
3.2.6. Protecting groups cleaved by β -elimination reactions

An important example of this subset is the 9-fluorenylmethoxycarbonyl (Fmoc) group.

It is commonly removed with secondary amines, such as piperidine, through a $E1cB$ mechanism. Piperidine removes the acidic proton at the C-9 position of the fluorene ring. This proton is acidic because the resulting anion is stabilized by the two aromatic rings. Dibenzofulvene is released, which reacts with piperidine forming a stable adduct that can be separated from the free amine.



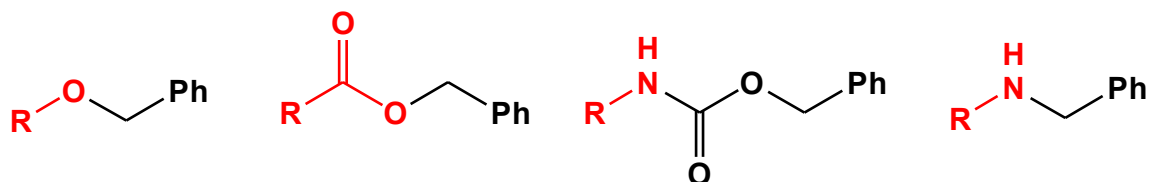
The 9-fluorenylmethyl (Fm) group is applied to the protection of carboxyl groups:



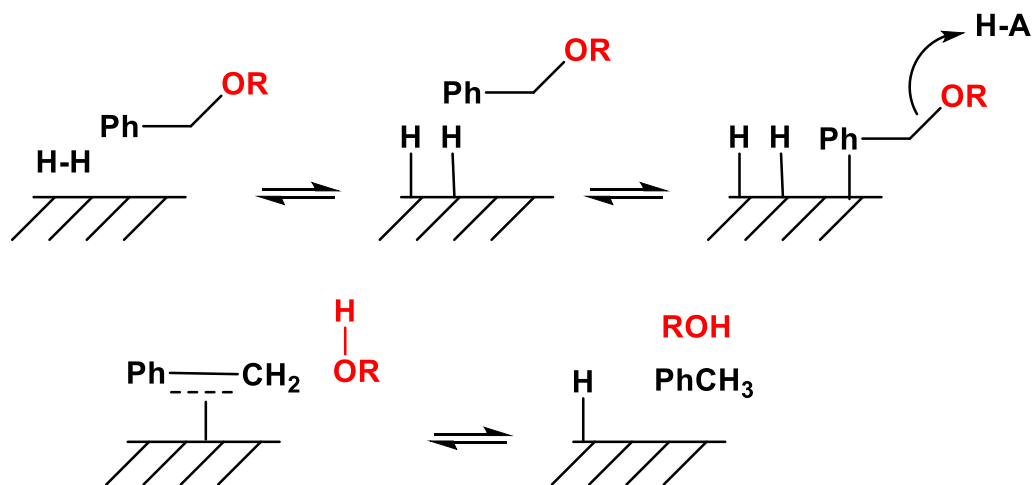
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3.2.7. Protecting groups cleaved by hydrogenolysis

An excellent method for cleaving benzylic ethers, esters, carbamates, and amines



It uses H_2 in the presence of a transition metal such as Pd

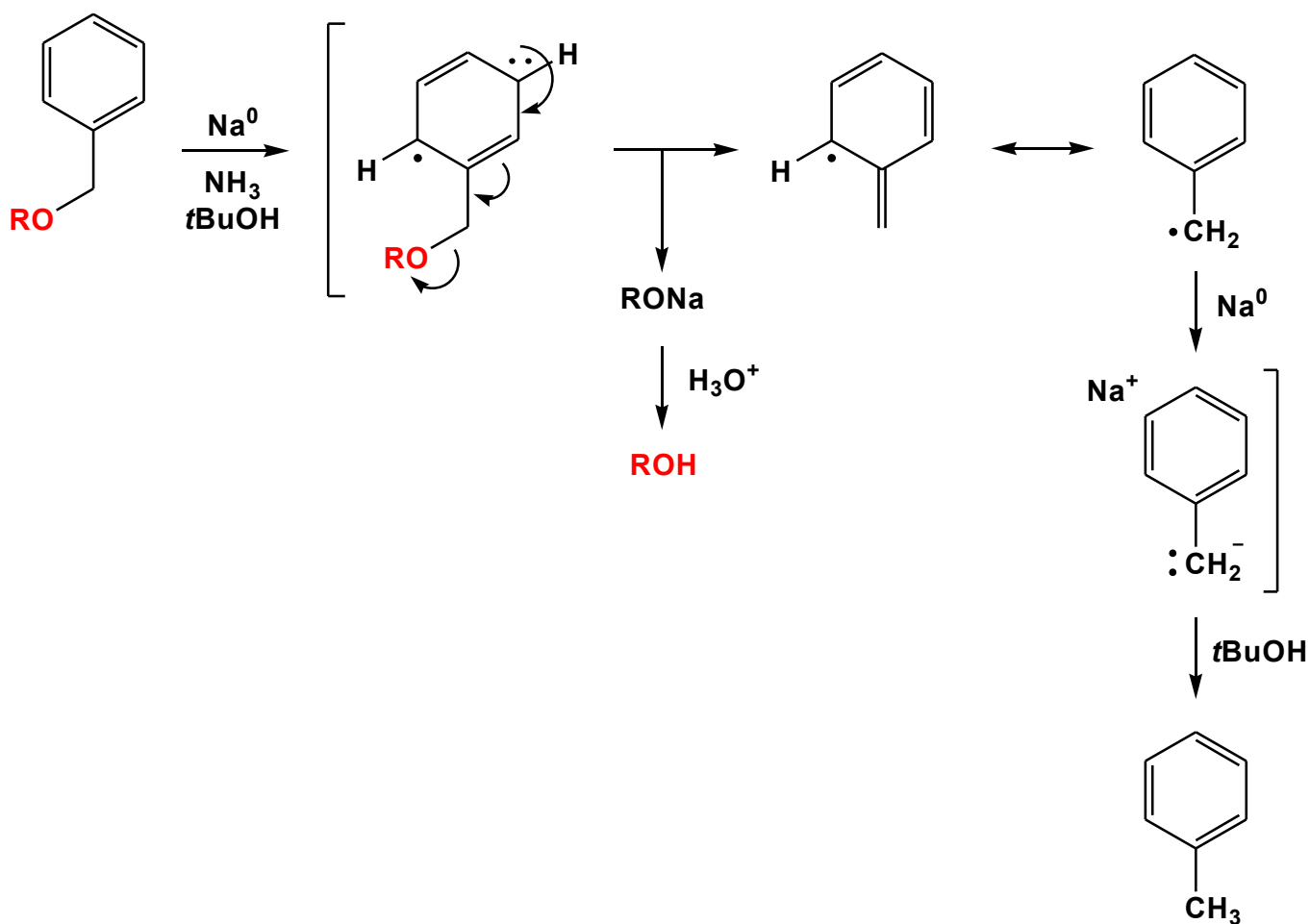


Alternatively a process known as *catalytic transfer hydrogenation* can be employed. It uses 1,4-cyclohexadiene, cyclohexene, formic acid or ammonium formate as H_2 source.

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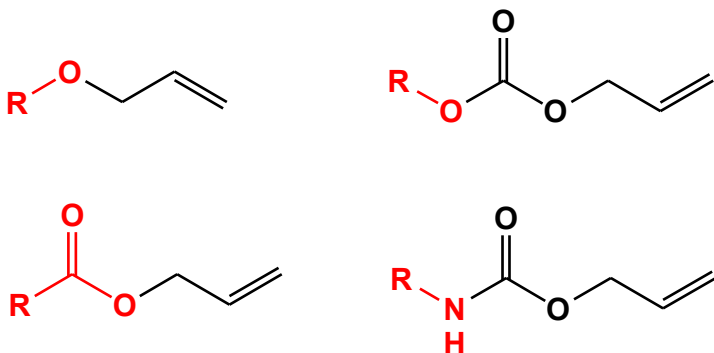
3.2.9. Protecting groups cleaved by dissolving metal reduction

Benzylic ethers and esters. Conditions: Na or Li in liquid NH_3

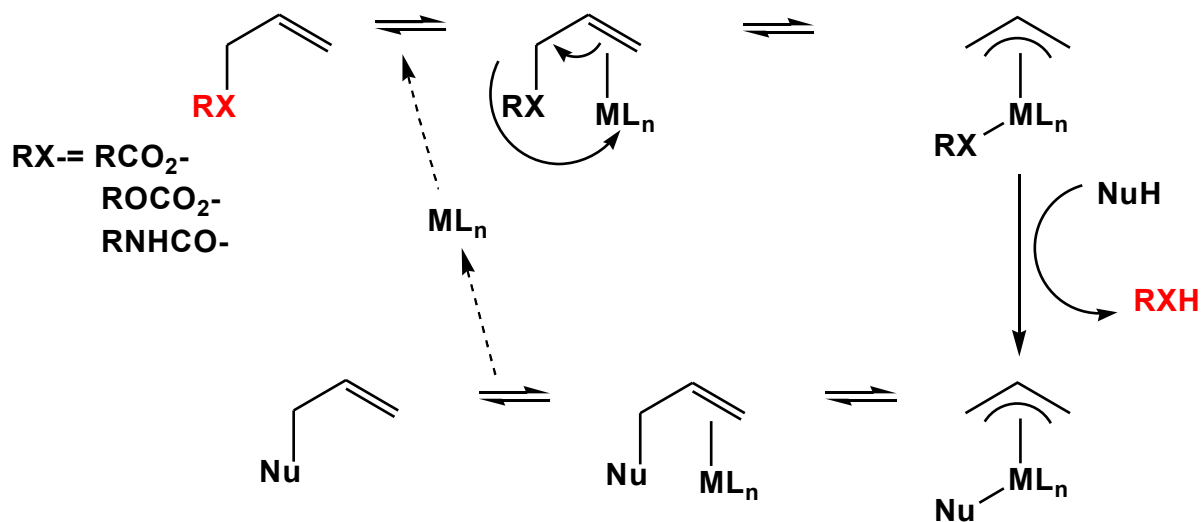


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3.2.10. Protecting groups cleaved by transition metal catalysis



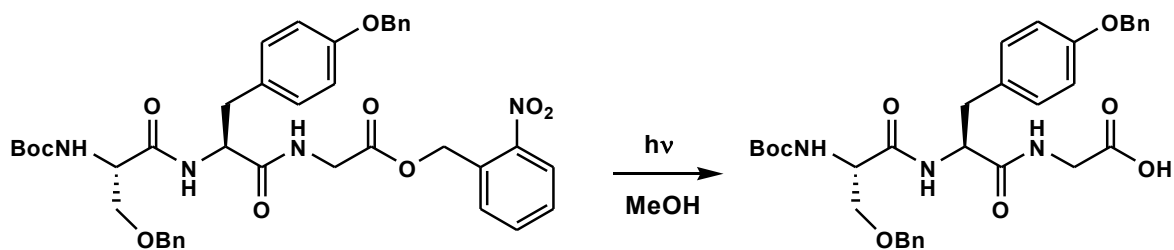
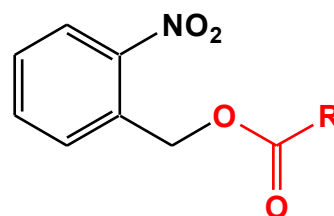
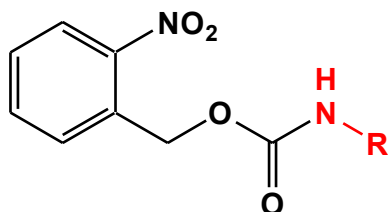
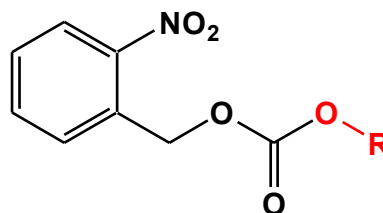
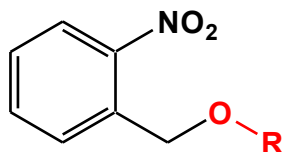
Conditions: $Pd(PPh_3)_4$, $PdCl_2$



NuH: Bu_3SnH , morpholine, AcOH

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3.2.11. Protecting groups cleaved by light



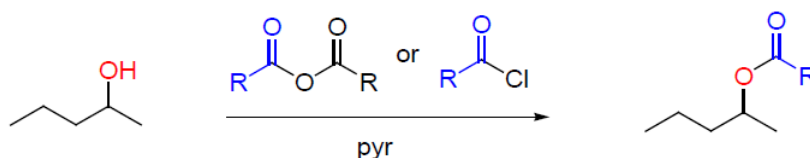
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3.3. Functional group protection

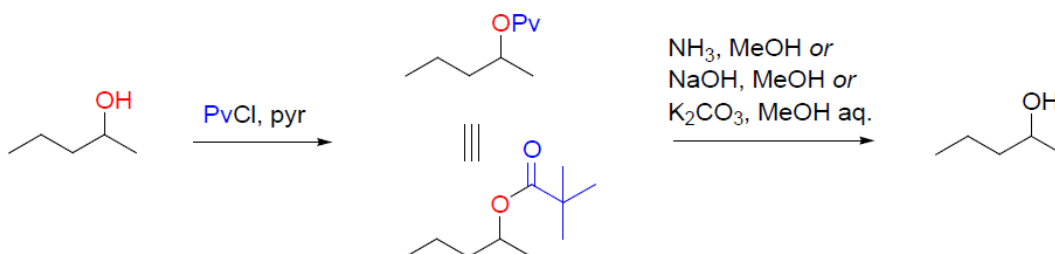
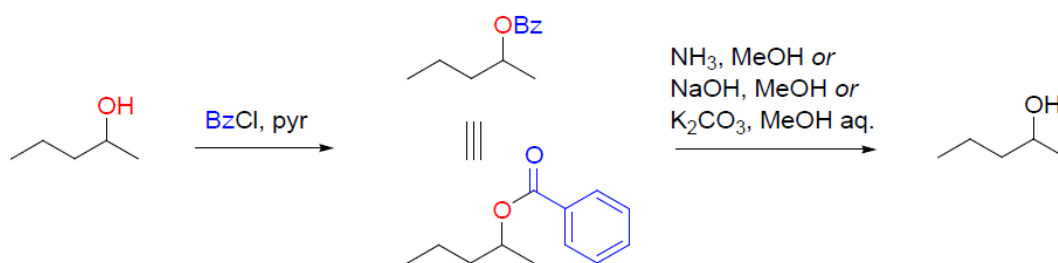
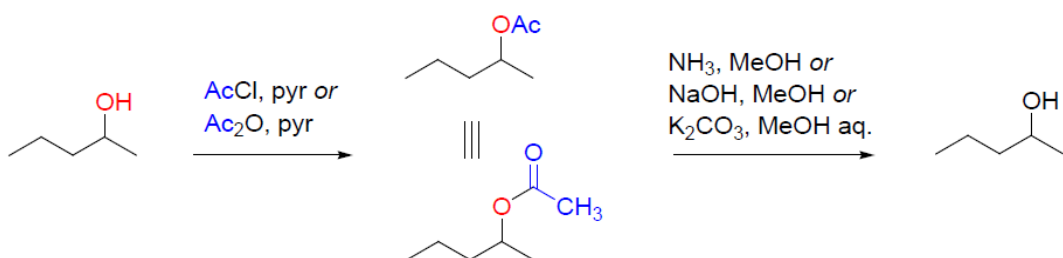
3.3.1. Hydroxyl protecting groups

Esters

Formation:



Examples and cleavage conditions:

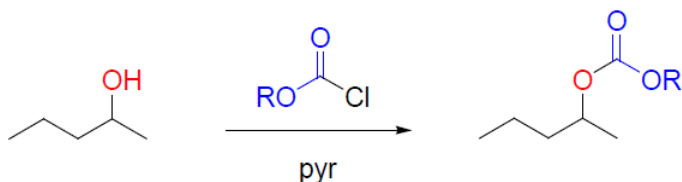


Reactivity order: Pv- < Bz- < Ac-

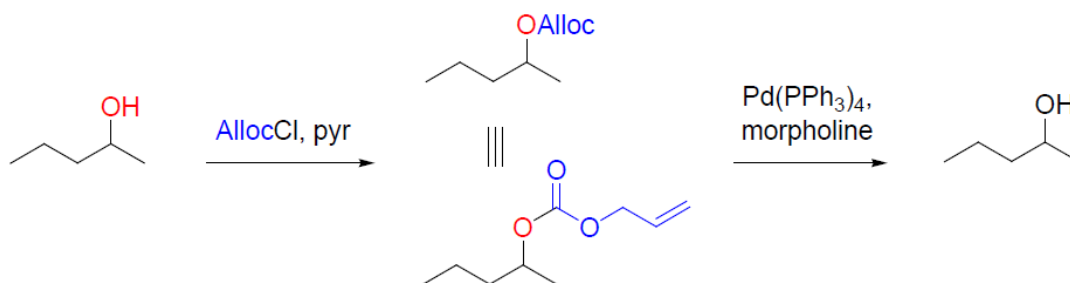
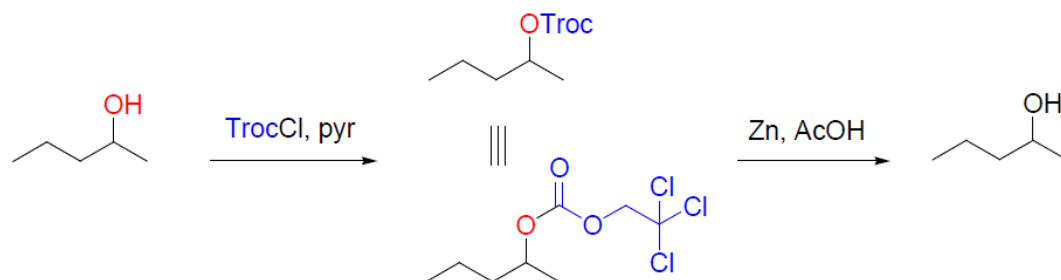
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Carbonates

Formation:

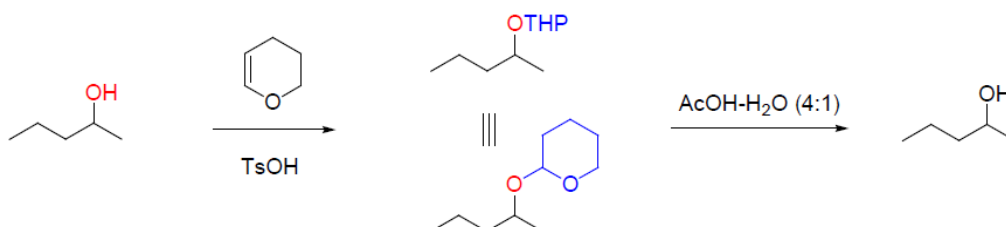


Examples and cleavage conditions:



Ethers

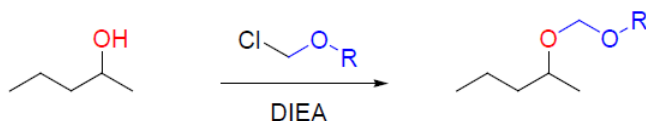
✓ Tetrahydropyranyl



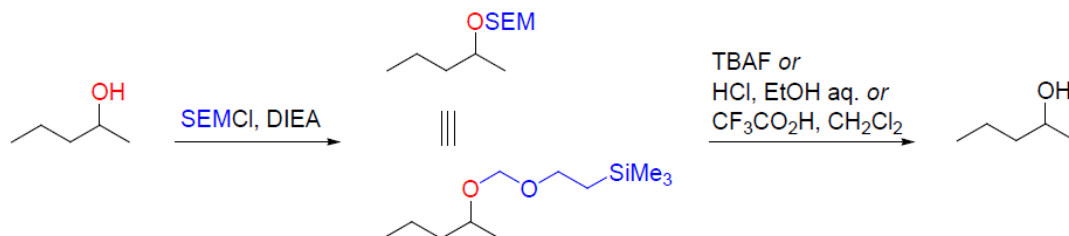
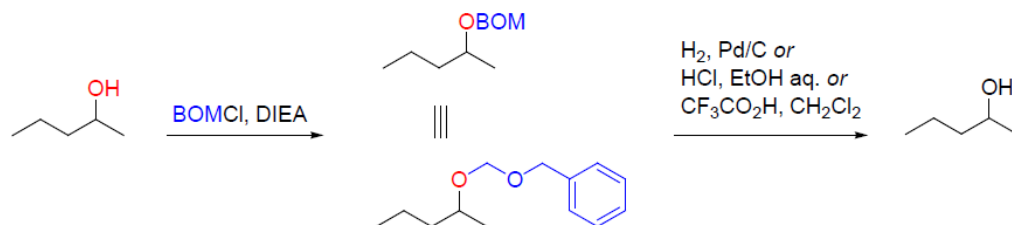
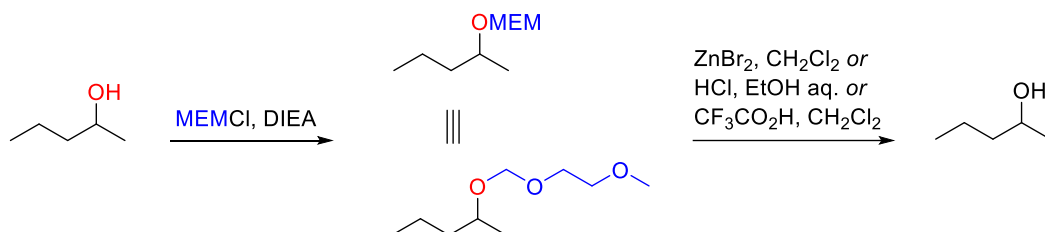
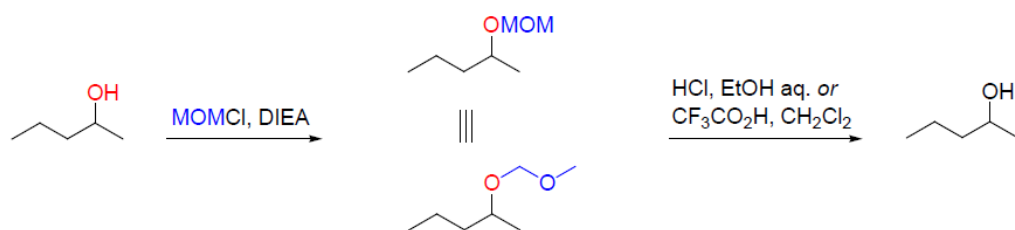
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✓ Alkoxymethyl ethers

Formation:

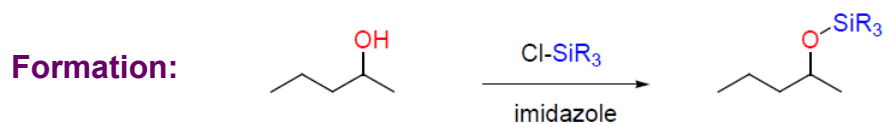


Examples and cleavage conditions:

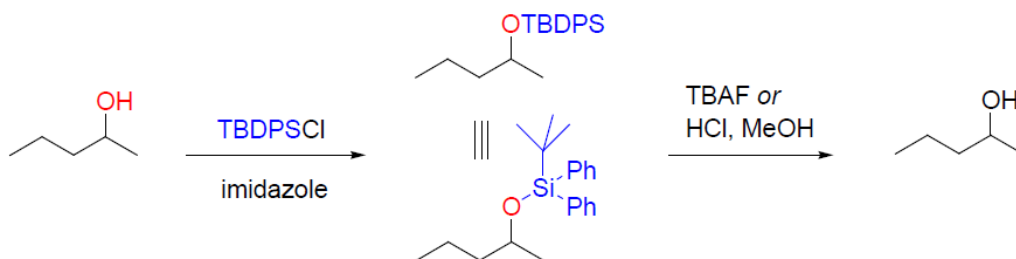
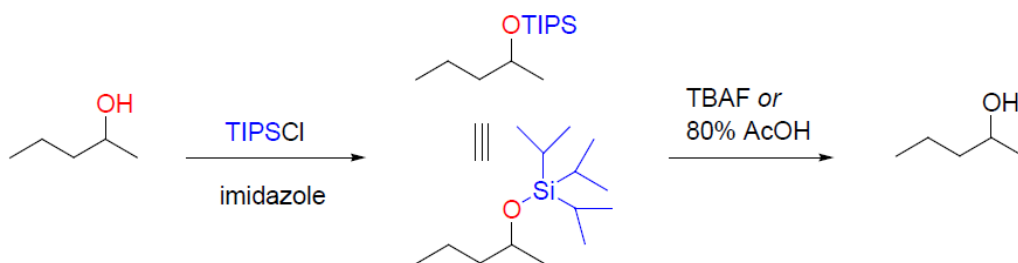
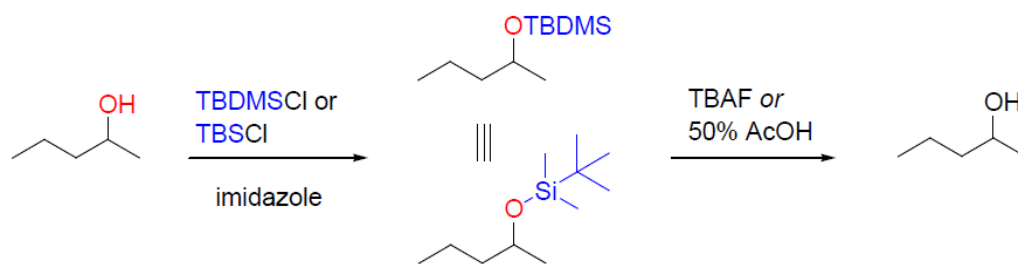


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✓ Silyl ethers



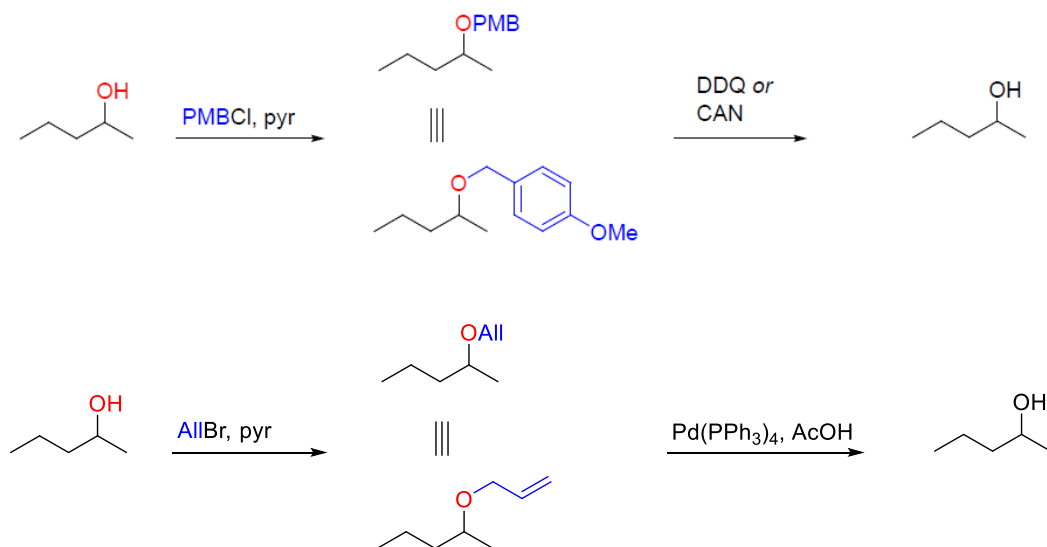
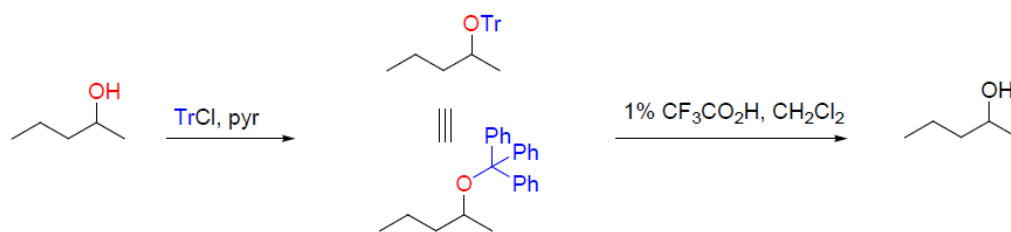
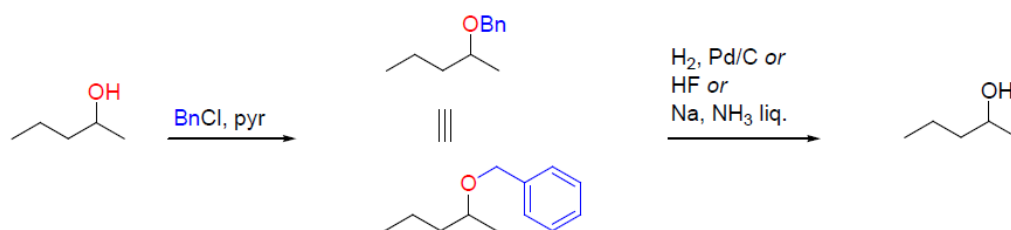
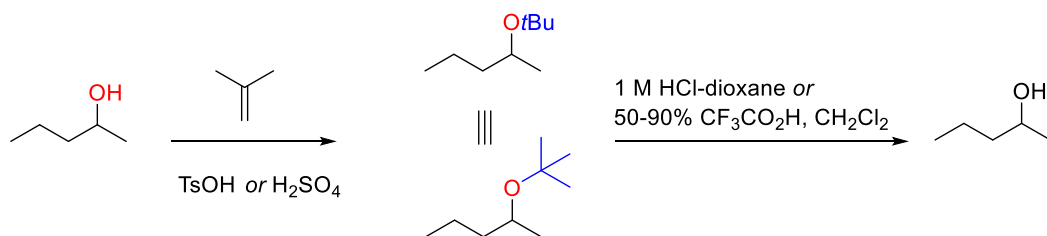
Examples and cleavage conditions:



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✓ Alkyl ethers

Examples, formation and cleavage conditions:

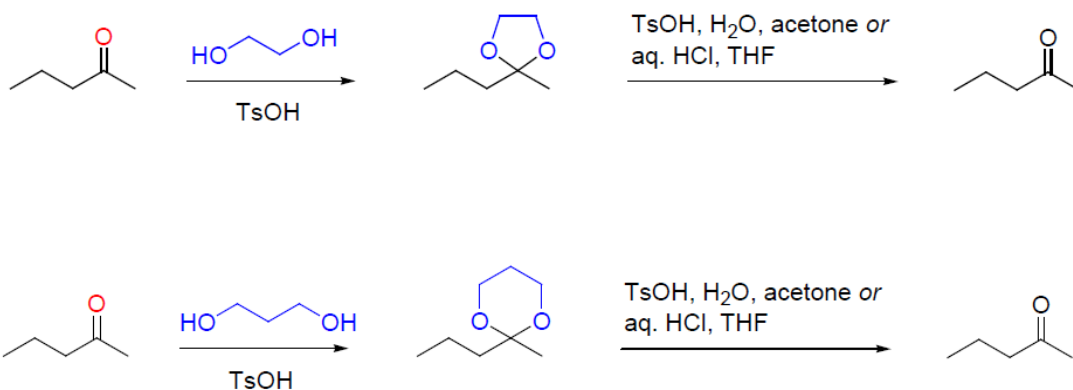


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3.3.2. Carbonyl protecting groups

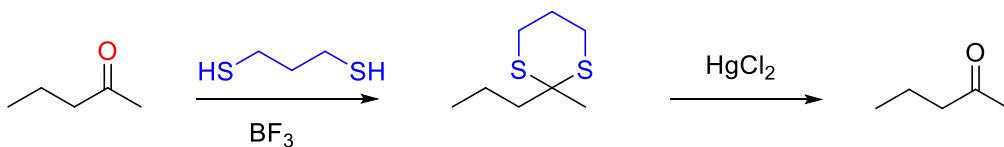
O,O-Acetals

Formation and cleavage conditions:



S,S-Acetals

Formation and cleavage conditions:

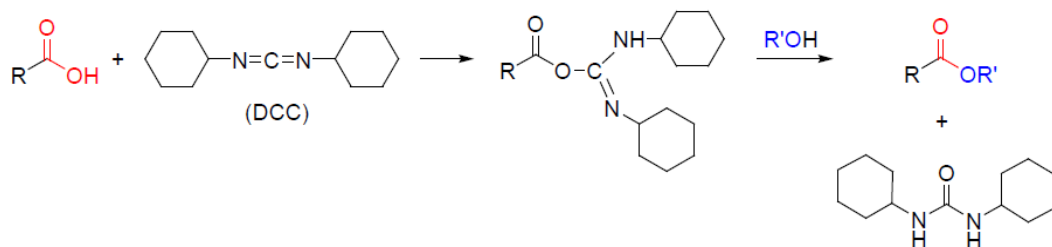
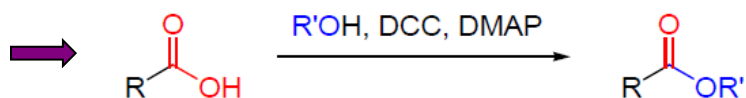
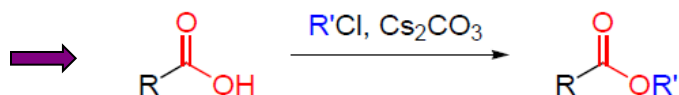
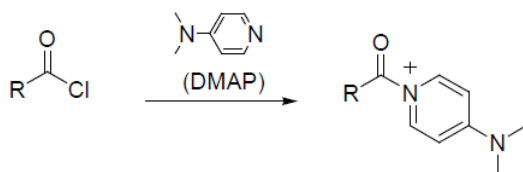
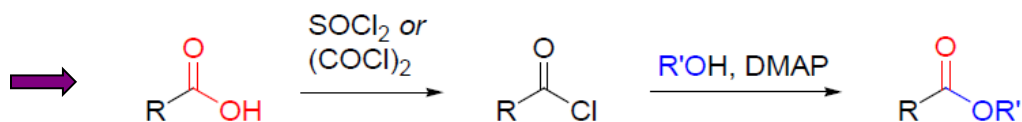
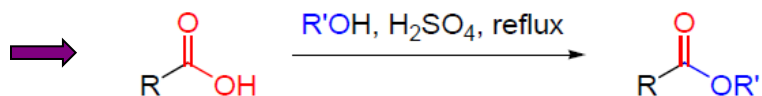


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3.3.3. Carboxyl protecting groups

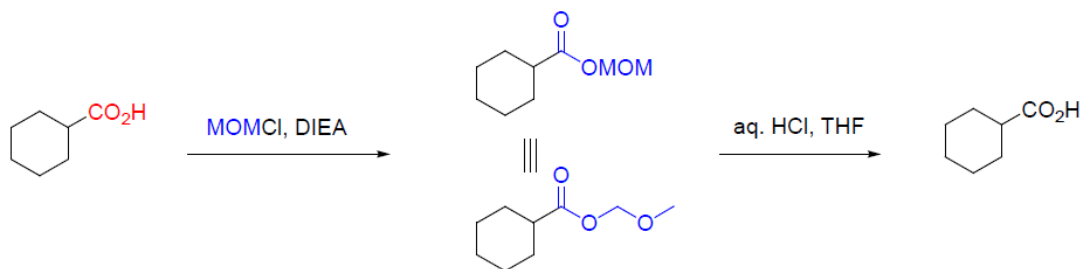
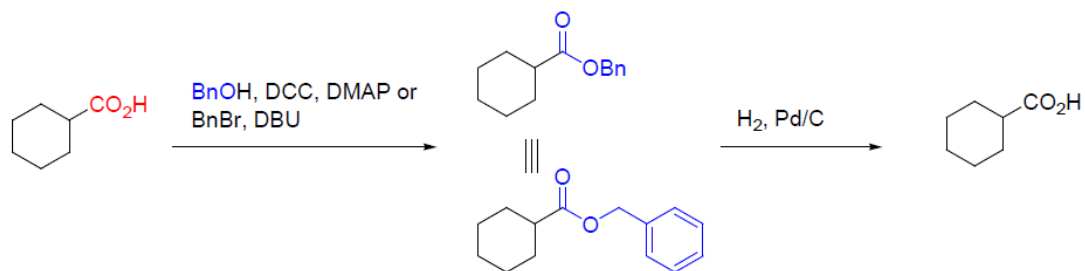
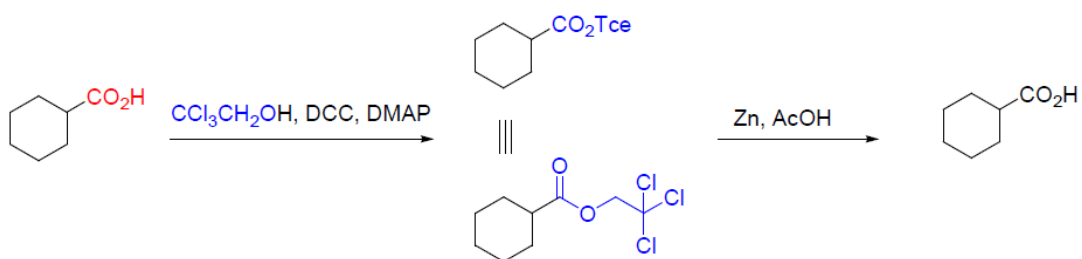
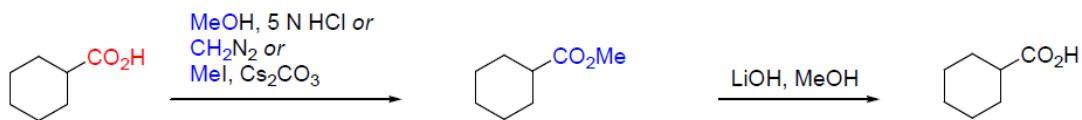
Esters

General esterification methods:

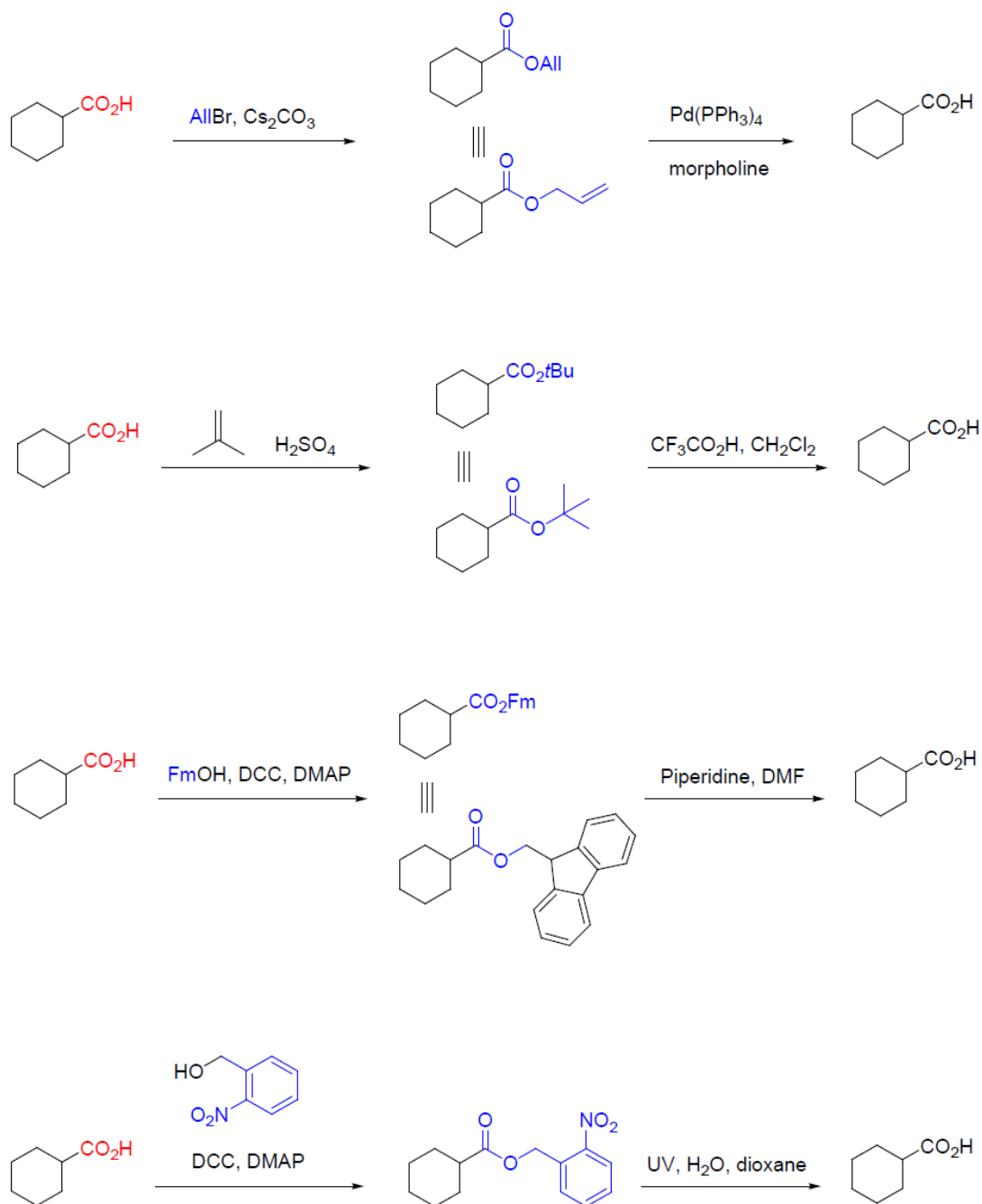


Chapter 3. Functional group protection

Examples, formation and cleavage conditions:



Chapter 3. Functional group protection



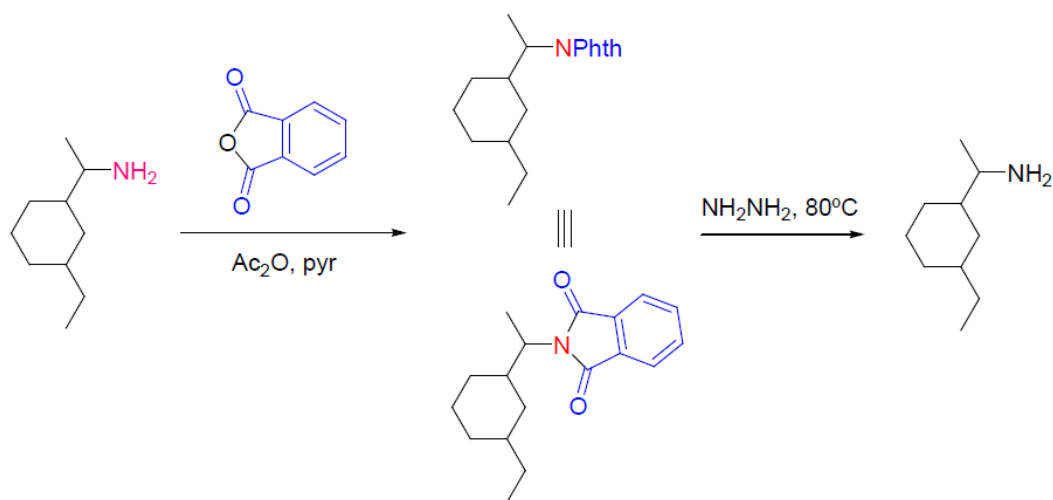
Chapter 3. Functional group protection

3.3.4. Amino protecting groups

Imides

Protecting groups for primary amines

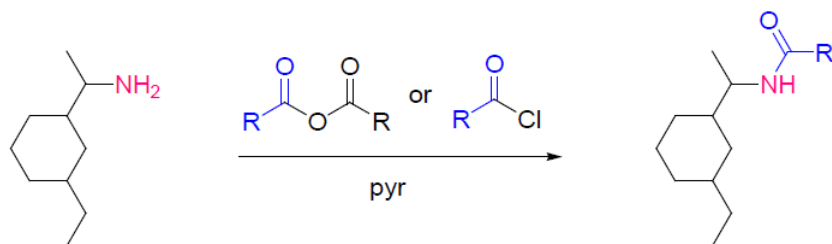
Formation and cleavage conditions:



Amides

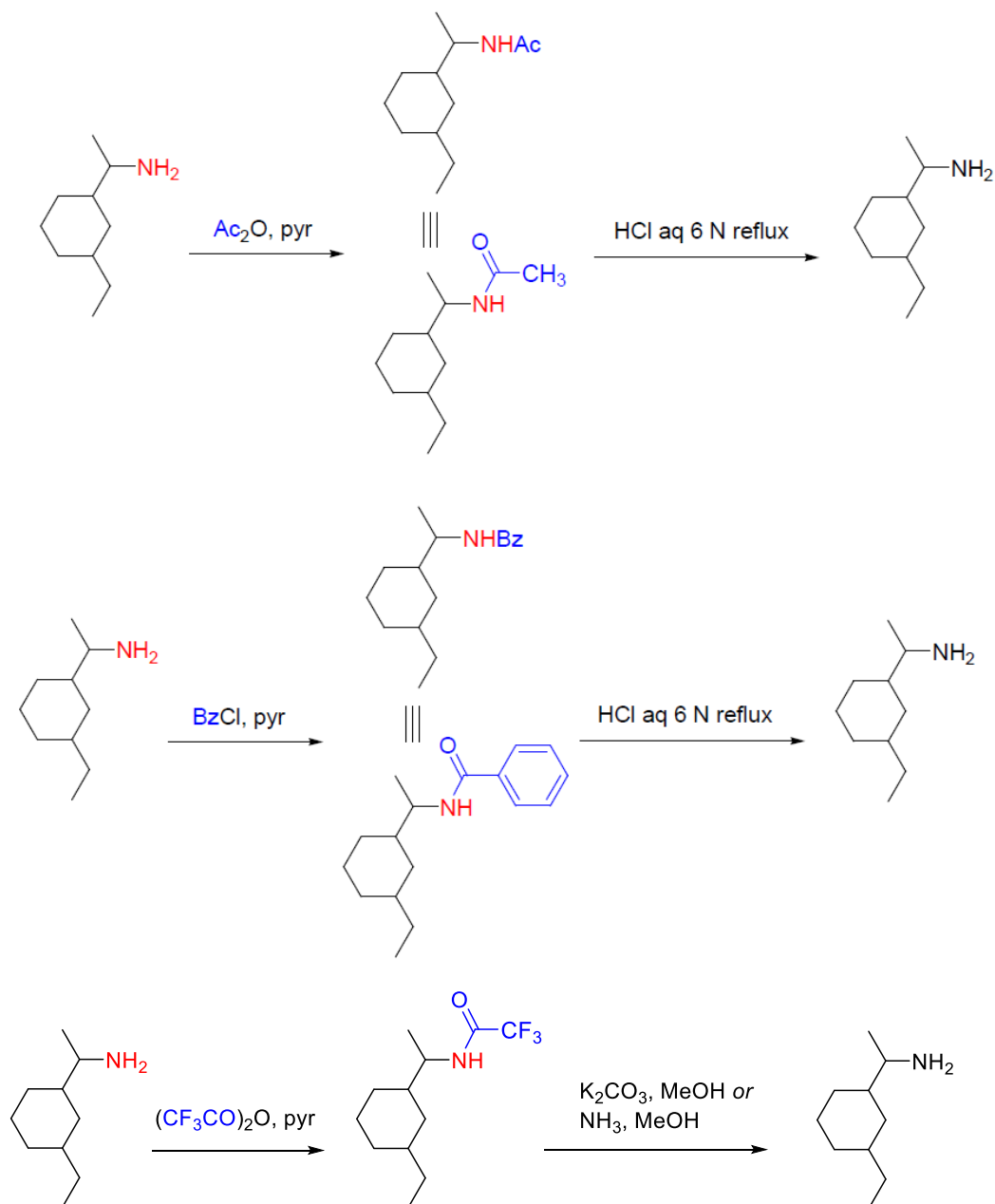
Protecting groups for primary and secondary amines

Formation:



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Examples and cleavage conditions:

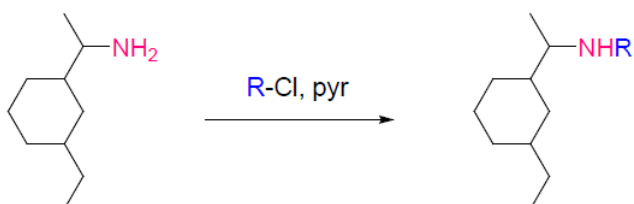


Chapter 3. Functional group protection

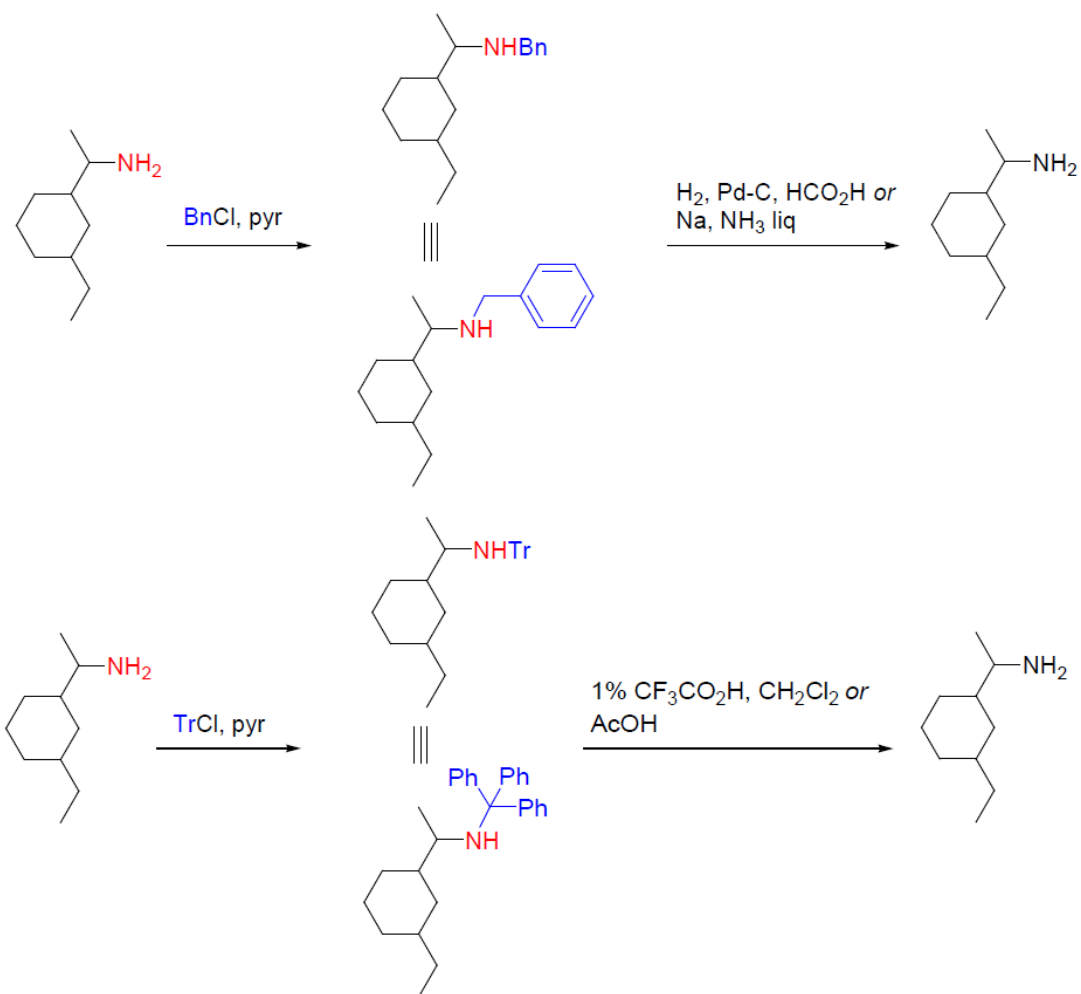
N-Alkyl derivatives

Protecting groups for primary and secondary amines

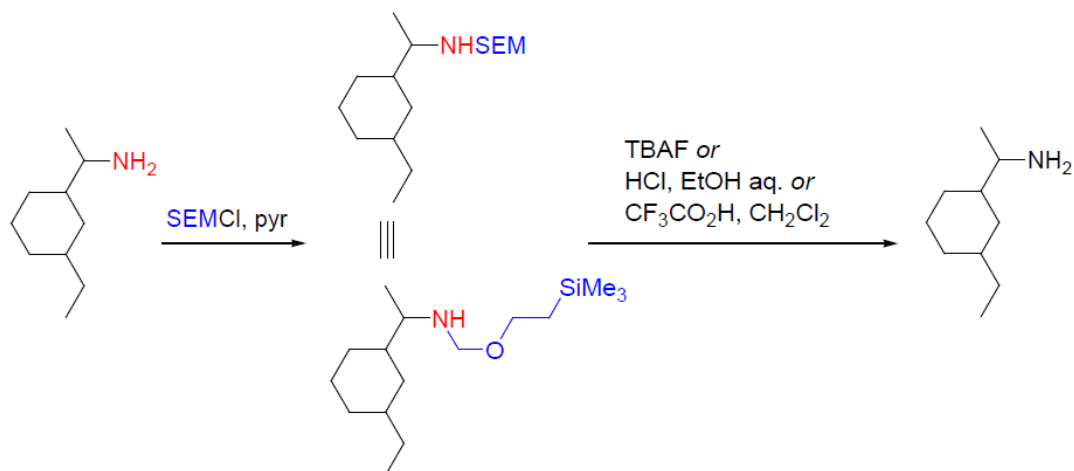
Formation:



Examples and cleavage conditions:



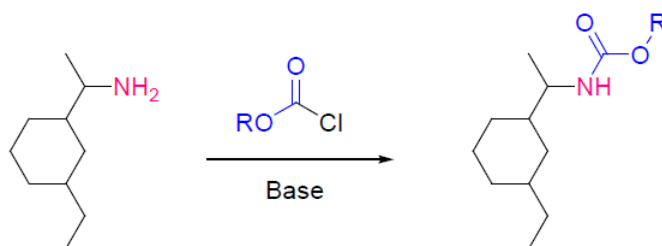
Chapter 3. Functional group protection



Carbamates

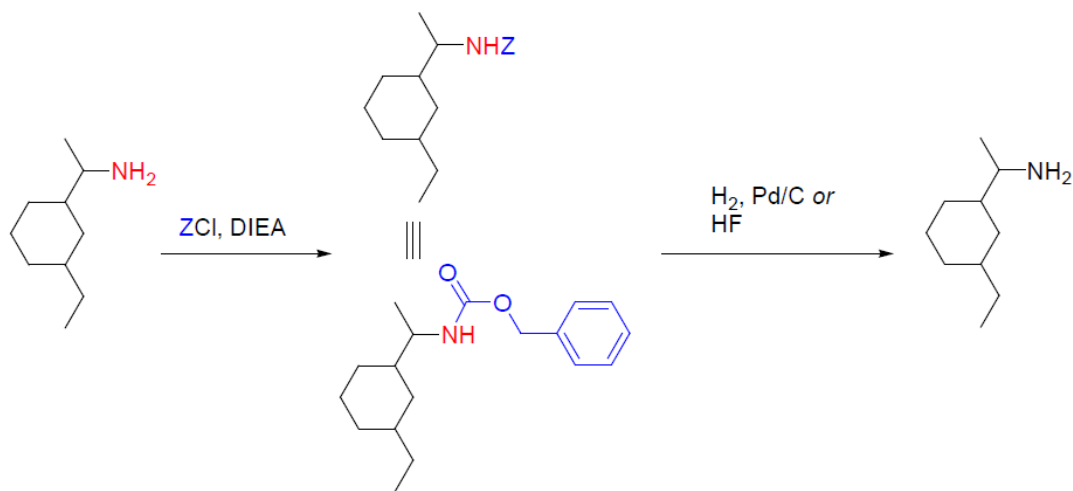
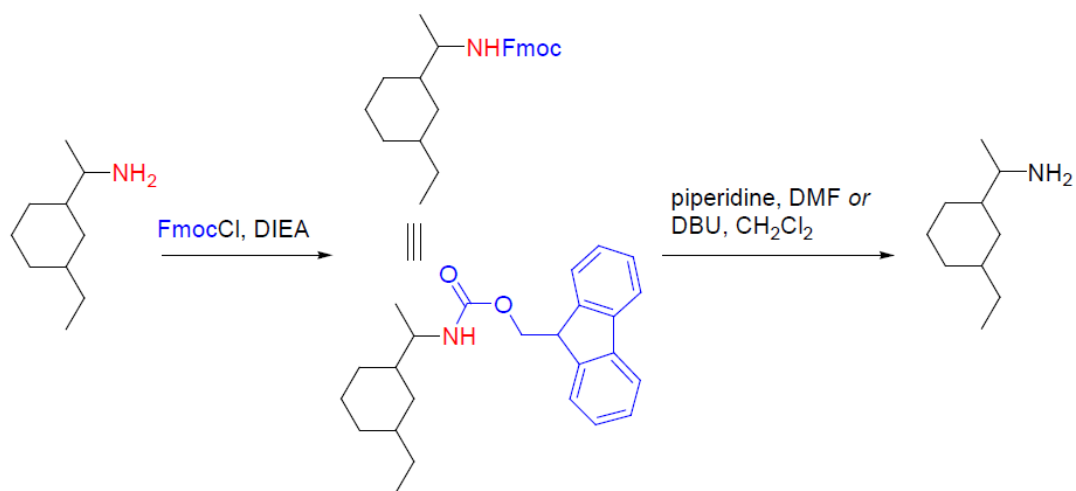
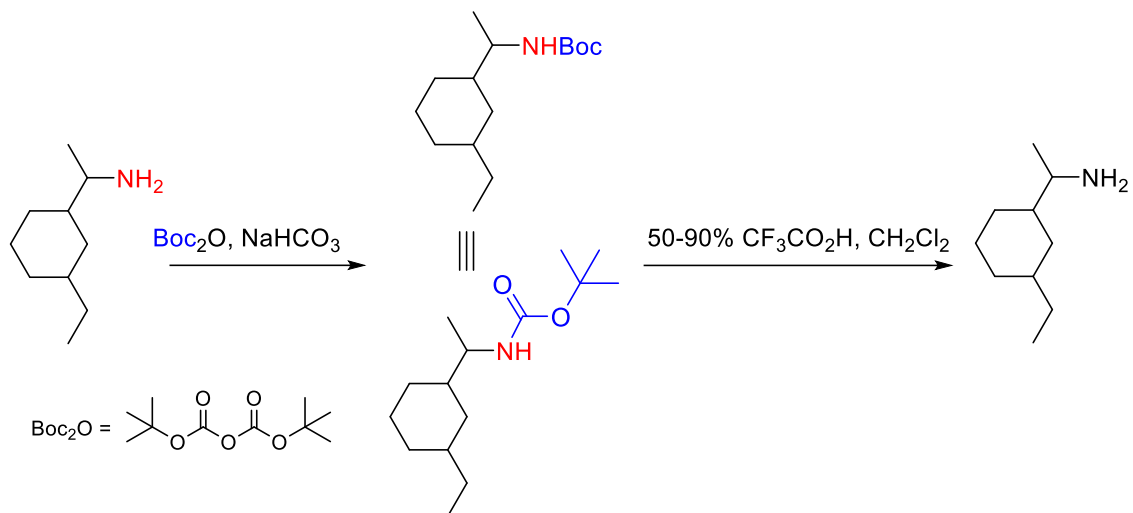
Protecting groups for primary and secondary amines

Formation:



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Examples and cleavage conditions:



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